

Firewall Configuration Theory TASK – 2

1. Install ufw on your Linux system (if not already installed) and enable it, then verify its status using the appropriate commands.

Ans :

- # Update package list
- sudo apt update

- # Install UFW
- sudo apt install ufw -y

- # Enable UFW
- sudo ufw enable

- # Verify UFW status
- sudo ufw status

- # Display detailed status (optional)
- sudo ufw status verbose

2. Configure ufw to allow incoming SSH connections only on port 22 and block all other incoming connections.

Ans :

- # Set default policies
- sudo ufw default deny incoming
- sudo ufw default allow outgoing

- # Allow SSH on port 22
- sudo ufw allow 22/tcp

- # Enable UFW
- sudo ufw enable

- # Verify the firewall rules
- sudo ufw status numbered

**3. Add a rule in ufw to allow outgoing HTTP and HTTPS traffic (ports 80 and 443), but block all other outbound connections.

Hint: Use 'ufw allow out' and 'ufw deny out' commands to control outbound traffic.**

Ans :

- # Set the default outbound policy to deny
- sudo ufw default deny outgoing
- # Allow outgoing HTTP traffic (port 80)
- sudo ufw allow out 80/tcp
- # Allow outgoing HTTPS traffic (port 443)
- sudo ufw allow out 443/tcp
- # Verify the firewall rules
- sudo ufw status numbered

4. List all currently active ufw rules on your system and save the output to a file named firewall_rules.txt.

Ans :

- # List all active UFW rules and save the output to a file
- sudo ufw status numbered > firewall_rules.txt
- # Verify the contents of the file

- `cat firewall_rules.txt`

- Status: active

To	Action	From
--	-----	----
22/tcp	ALLOW IN	Anywhere
80/tcp	ALLOW OUT	Anywhere
443/tcp	ALLOW OUT	Anywhere
22/tcp (v6)	ALLOW IN	Anywhere (v6)
80/tcp (v6)	ALLOW OUT	Anywhere (v6)
443/tcp (v6)	ALLOW OUT	Anywhere (v6)

5. Write a short explanation (3-4 lines) describing how your current ufw configuration would impact running a local web server (like a development server for a Zomato-style food delivery app).

Ans : The current UFW configuration allows only incoming SSH connections on port 22 and blocks all other incoming traffic. As a result, a local web server running on ports such as 80, 443, or 3000 would not be accessible from other devices on the network. The server can still make outbound HTTP and HTTPS requests, but to access the web application externally, you would need to add a UFW rule allowing the web server's port (for example, `sudo ufw allow 80/tcp` or `sudo ufw allow 3000/tcp`).